

8-BIT ALU DESIGN USING VERILOG HDL

Internship Project Report

Submitted By

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Introduction

What is 8-bit ALU?

An Arithmetic Logic Unit (ALU) is one of the most important components of a digital system and processor, it is the computational core of any processor. It performs arithmetic and logical operations on binary data. 8-bit ALU processes operands that are 8 bits wide, making it a foundational building block in digital design education.

Importance of ALU

- Performs arithmetic calculations.
- Executes logical operations.
- Used in CPUs and microprocessors.
- Essential component in digital systems.

This project presents the design and verification of an 8-Bit ALU using Verilog HDL. The ALU performs arithmetic, logical, and shift operations based on a 3-bit select signal. Functional verification was carried out using Xilinx Vivado simulation.

Objective

The objectives of this project are:

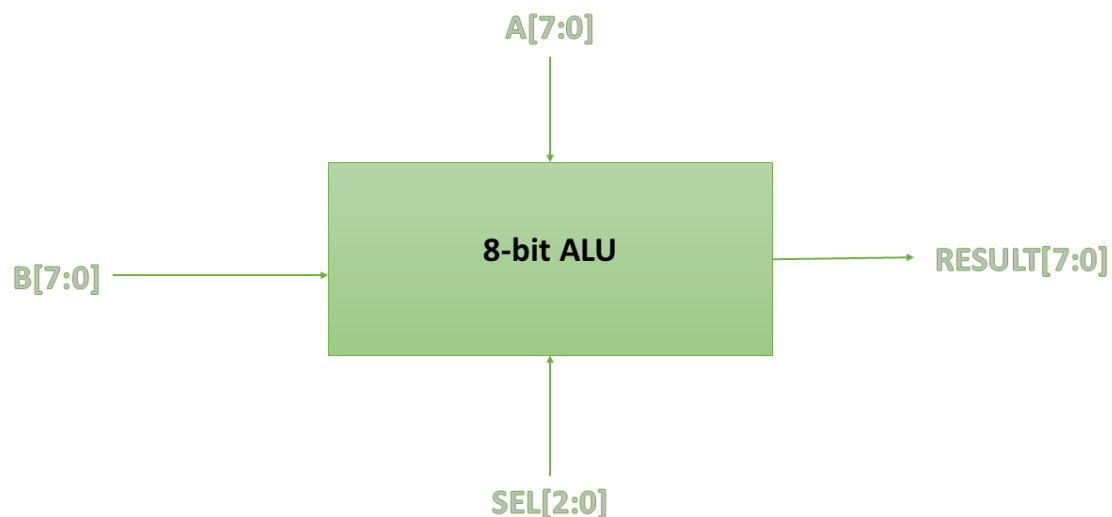
- Design an 8-Bit ALU using Verilog HDL.
- Implement arithmetic operations.

- Implement logical operations.
- Implement shift operations.
- Verify functionality using simulation.
- Generate RTL representation.

Tools used-

- Verilog HDL
- Xilinx Vivado 2025.2
- GitHub
- Windows 11

Block Diagram



The ALU consists of two 8-bit inputs A and B, a 3-bit select signal SEL, and one 8-bit output RESULT. Depending on the value of SEL, the required operation is performed.

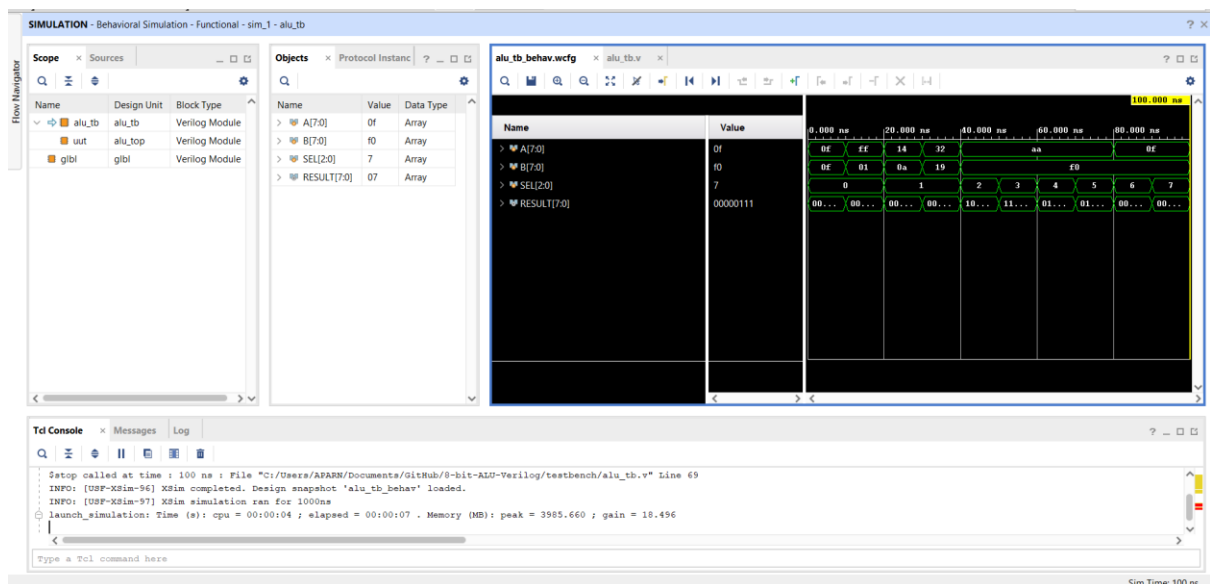
Verilog Modules

Modules used:

- adder.v
- subtractor.v
- and_gate.v
- or_gate.v
- xor_gate.v
- not_gate.v
- shifter.v
- alu_top.v

The top module integrates all submodules and selects the appropriate output.

Testbench

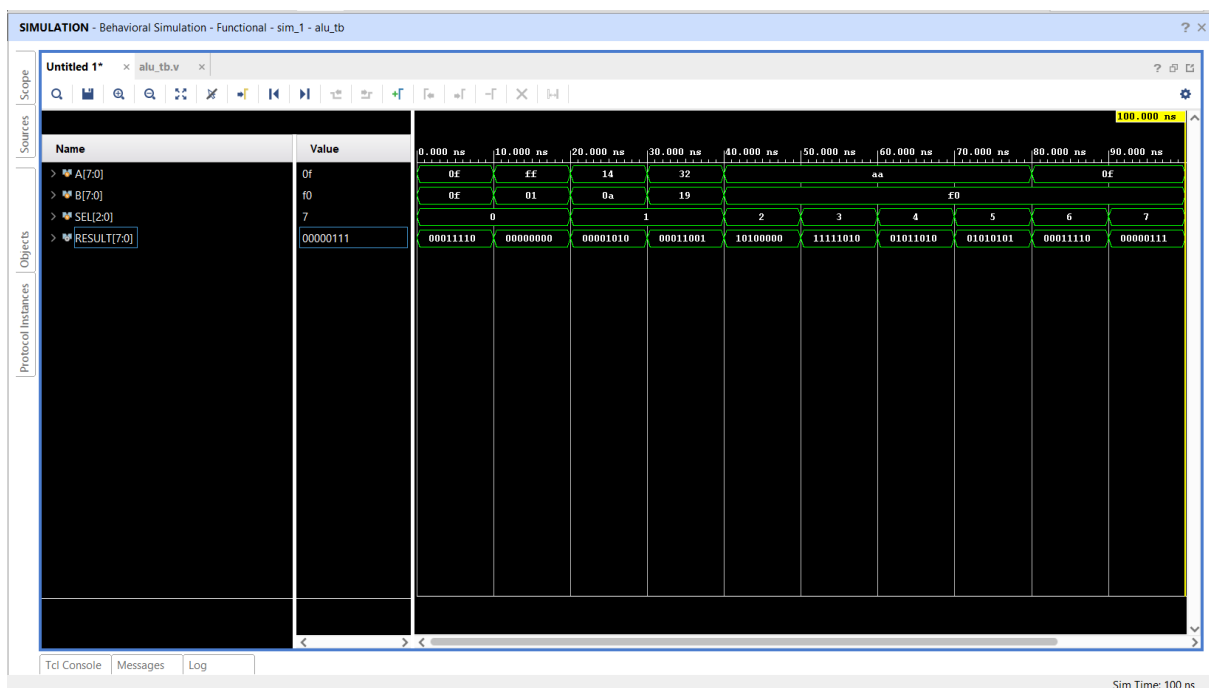


A testbench was developed to verify all ALU operations.

Test Cases-

Test No.	Operation
1	ADD
2	ADD Overflow
3	SUB
4	SUB
5	AND
6	OR
7	XOR
8	NOT
9	LEFT SHIFT
10	RIGHT SHIFT

Waveforms

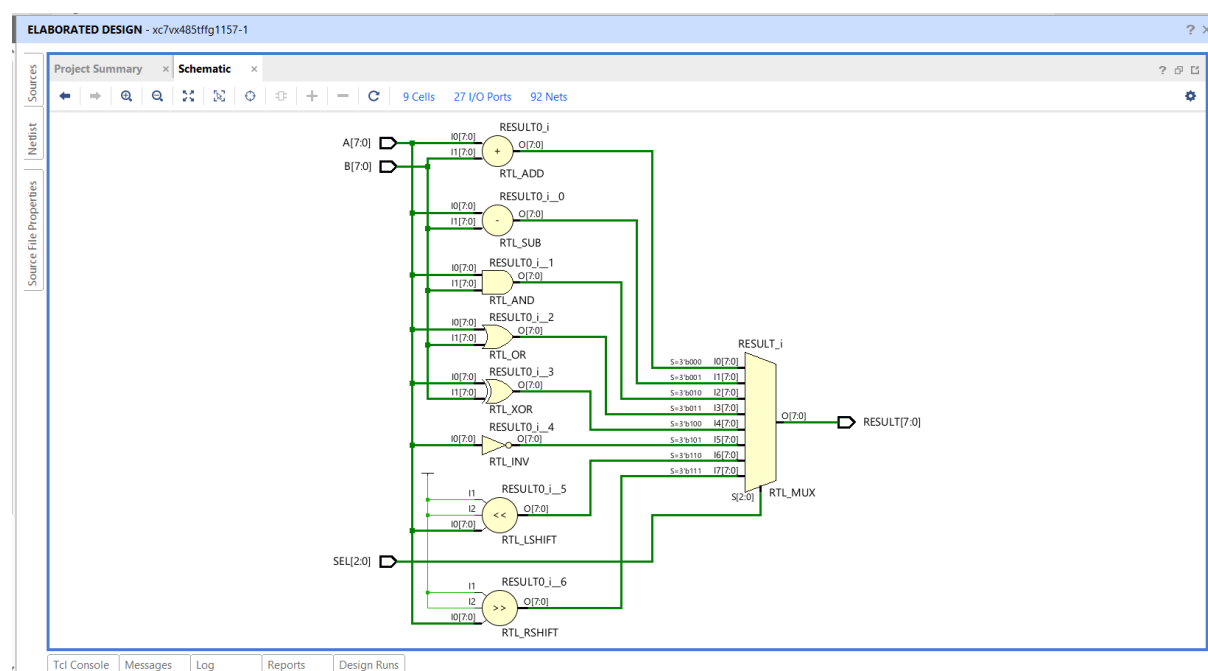


The ALU was simulated in Xilinx Vivado.

Observations

- All arithmetic operations produced expected outputs.
- Logical operations were verified successfully.
- Shift operations worked correctly.

RTL Schematic



The RTL schematic represents the hardware structure generated from the Verilog code before synthesis.

Conclusion

The 8-Bit ALU was successfully designed and verified using Verilog HDL. The project demonstrated arithmetic, logical, and shift operations controlled through a 3-bit select signal. The ALU successfully performed all eight operations. Simulation outputs matched the expected values for all test vectors and the Simulation results confirmed correct functionality.